

SIFATULLAH SHEIKH

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PROFESSIONAL SUMMARY

Computer Science graduate with one year of research experience in machine learning and deep learning, focusing on computer vision, multimodal AI, and Trustworthy AI. Conducted research in computer vision and developed intelligent systems for real-world applications in healthcare and biodiversity. Interested in advancing research at the intersection of reinforcement learning, AI-enabled autonomy, computer vision, robotics, and multi-agent systems, with the long-term goal of developing safe, reliable, and interpretable AI systems for autonomous decision-making in real-world environments.

EDUCATION

B.Sc. in Computer Science and Engineering (CGPA: 3.70/4.00)

East West University

Dhaka, Bangladesh

Jan 2022 – Jan 2026

Relevant Coursework: Machine Learning, IOT, Computer Vision, Database Systems, Software Engineering, Data Structures & Algorithms, Data Mining.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant

Dhaka, Bangladesh

East West University

Dec 2024 – Dec 2025

- Benchmarked four deep learning architectures (three CNNs and one Vision Transformer) for AI-generated face detection using a dataset of 140K real and AI-generated face images, identifying complementary strengths between CNNs for local feature extraction and transformers for global contextual understanding.
- Designed a cross-attention fusion architecture that combines Swin Transformer's global reasoning with EfficientNet's local feature extraction using a multi-head cross-attention mechanism, addressing the complementary limitations of individual architectures.
- Conducted ablation studies and evaluated the proposed framework using ROC-AUC, log loss, Brier score, and Cohen's Kappa to assess model robustness and quantify the contribution of each architectural component.
- Integrated Explainable AI techniques (Grad-CAM and LIME) to interpret model predictions and evaluated the framework on unseen datasets to assess its generalization across different data distributions.

TECHNICAL SKILLS

Programming: Python, C++, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face Transformers

Research: Computer Vision, Multimodal AI, Explainable AI (XAI), Statistical Analysis, Ablation Studies, LaTeX

Development Tools: Git, GitHub, Docker, Google Colab

Hardware & Embedded: Raspberry Pi, Arduino, Sensor Integration, Embedded Systems

Backend: Flask, FastAPI

PUBLICATIONS

- [J1] DeFaX: A Cross-Attention Fusion Framework for Robust and Explainable Deepfake Detection
Md Al-Imran, **Md Sifatullah Sheikh**, Urmi Kirtonia, Nuzath Tabassum Arthi, Shamim Ripon
Top-tier Computer Science journal, 2025. ieeexplore.ieee.org/abstract/document/11303744
- [J2] Medicinal Plant Leaf Image Dataset of 13 Species Collected in Bangladesh
Momena Khatun Zinia, Mahmudul Haque Sakib, **Md Sifatullah Sheikh**, Urmi Kirtonia, Bashirul Islam, Md Nawab Yousuf Ali
Scientific Data (Nature Portfolio Journal).
- [C1] AI-Powered Deepfake Detection Using CNN and Vision Transformer Architectures
Md Sifatullah Sheikh, Urmi Kirtonia, Nuzath Tabassum Arthi, Md Al-Imran
IBDAP International Conference, Thailand, 2025. ieeexplore.ieee.org/abstract/document/11145852
- Dataset: A Medicinal Plant Leaf Image Dataset of 13 Species Collected in Bangladesh – Momena Khatun Zinia, Mahmudul Haque Sakib, **Md Sifatullah Sheikh**, Urmi Kirtonia, Bashirul Islam, Md Nawab Yousuf Ali. *Mendeley Data*, Version 2, 2026.
data.mendeley.com/datasets/9tdc9gbtgb/2

PROJECTS

MediLeafNET – Multimodal Medicinal Plant Identification

Undergraduate Thesis, 2025

PyTorch, ViT, BERT, OpenCV, Gradio, Flutter

github.com/SifatSwapnil2022/MediLeafNET

- Built a multimodal classifier on 3,289 leaf images (data collected by our group), reaching 96.22% accuracy and outperforming single-modal baselines by 8%.
- Deployed a Gradio demo and integrated with Flutter for a mobile proof-of-concept.

SkinCare AI – Skin Disease Detector with LLM Advisor

2025

YOLOv8, ResNet50, MobileNetV2, EfficientNetB0, FastAPI, Streamlit, Docker

github.com/SifatSwapnil2022/SkinCareAI

- Engineered an end-to-end pipeline: lesion detection (YOLOv8, 95.42% mAP) into ensemble classification (ResNet50 93.11%, EfficientNetB0 91.57%, MobileNetV2 89.80%) into LLM-generated recommendations.
- Balanced the dataset to 7,000 images per class across 10 categories.

AWARDS & RECOGNITION

- Dean's List Scholarship (3 consecutive semesters) – East West University 2023–2025
- 3rd Place, East West University National RoboFest 2024
- Finalist, ICT Olympiad Bangladesh 2026 (National Programming & ICT Competition) ictolympiadbangladesh.com/winners

EXTRACURRICULAR & VOLUNTEER

- **Executive Member**, East West University Robotics Club
Organized events and workshops.
- **Core Member**, LMH Foundation
Participated in food and clothing drives for underprivileged communities. facebook.com/profile.php?id=100064420736877